ACADEMIC PROJECT DEFENSE

LIBRARY MANAGEMENT SYSTEM

A terminal-based system for smarter library operations

BRAINWARE UNIVERSITY

Department of Computational Sciences

Presented by

Rahul Das

Student code

BWU/MCA/26/021

Registration no.

26102000139

Roll no.

26102410017

Course: Problem Solving using Python · Course Code: MCA17400(T)

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SYSTEM OVERVIEW

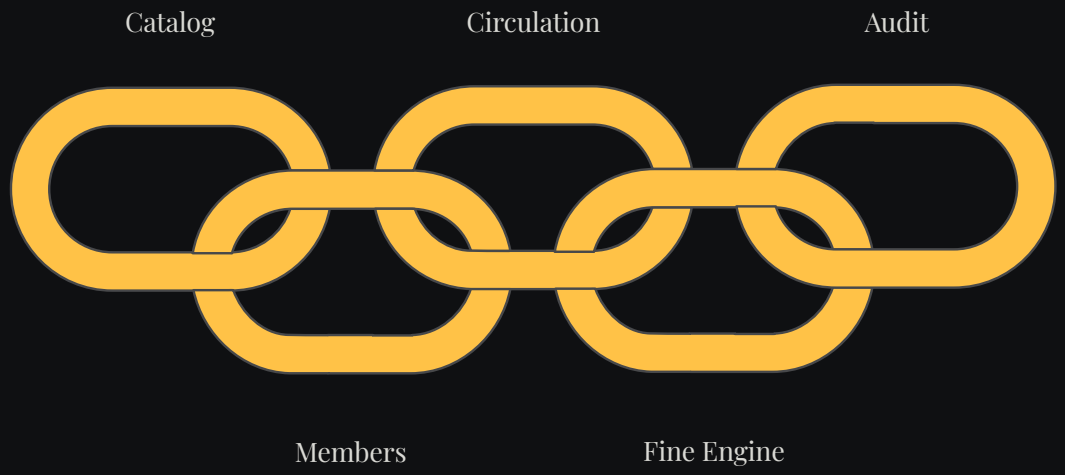
Library Management System

A terminal-based system for smarter library operations

Replace blind ID entry and fragmented records with visible, validated workflows.

Object-oriented Python application for catalog, members, circulation, overdue fines, and financial auditing.

Built with Python OOP and embedded SQLite3 persistence.



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INTERACTIVE LIFECYCLE

The system works as one interactive circulation loop

Preview first. Commit the transaction. Track the state. Audit the result.

01

Preview

Preview applicable books, students, active issues, or dues before asking for an ID.

02

Issue / return

Update stock and issue records transactionally.

03

Fine state

Calculate overdue charges at ₹5.00 per late day and move the fine through NONE, DUE, or PAID.

04

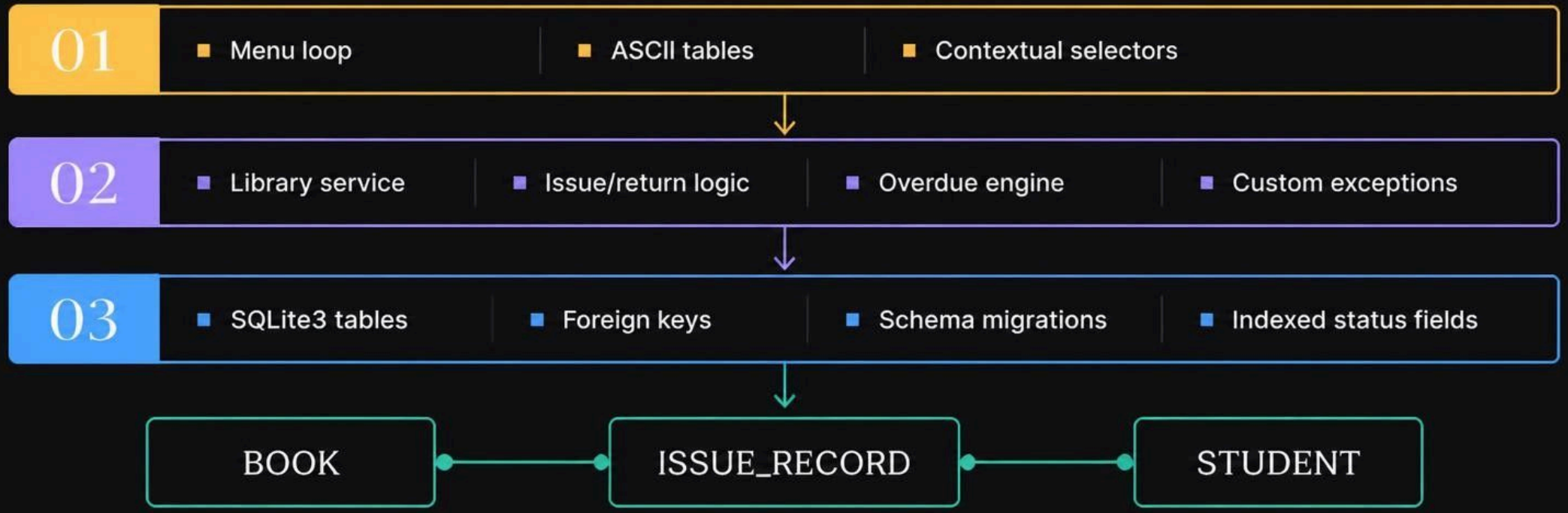
Audit

Audit outstanding dues, collected income, active loans, and borrowing statistics.

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ARCHITECTURE + DATA MODEL

Three layers keep the rules clear and the data safe



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RELIABILITY EVIDENCE

Reliability is visible in the edge cases and the evidence

TEST RESULT

54/54

unit tests passed

Across 7 focused test classes.

Edge cases

Stock depletion · duplicate loans · invalid inputs · schema changes · date boundaries

Benchmark

0.154 seconds for the full test suite

Relational joins

under 2 ms

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EXTENSIBILITY

From academic project to extensible platform

CURRENT RESULT

Focused and dependable

A lightweight, zero-dependency library automation system with OOP modeling, SQLite3 persistence, contextual previews, and financial state management.

FUTURE SCOPE

Ready to extend

Barcode/RFID scanning, automated overdue email reminders, and a REST API with a graphical dashboard.

Key takeaway: dependable library operations can be built with a focused terminal interface and disciplined architecture.

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LIBRARY MANAGEMENT SYSTEM

Thank you

Open for questions

Rahul Das · MCA · Brainware University

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